10 Sovereignmemory

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Sovereign Memory Benchmark — JS

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1. Performance

Operation	Time
remember() (write+encrypt)	8.2ms
recall() (decrypt+lookup)	3.1ms
$\text{search}_{\text{text}}$ (1K entries)	4.8ms
$\text{search}_{\text{semantic}}$ (1K entries, 384-dim)	12ms
Reload from disk (10K entries)	340ms

2. Encryption

Property	Value
Algorithm	AES-256-GCM
Key derivation	scrypt (N=16384, r=8, p=1)
Nonce	96-bit random per write
Auth tag	128-bit GCM
Password → key time	~200ms (intentionally slow)

3. Storage

Each entry: ~500 bytes average (compressed JSON + encryption overhead).
10,000 entries ≈ 5MB. Unlimited retention — local disk only.

4. Data Sovereignty

- Zero cloud sync
- Zero telemetry
- Works fully offline forever
- Password known only to deploying organization